Hardware User Manual

MLII Series

ML-II-DB96T

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Revision History

Manual	PCB	Comments
Rev. 1.0 Issue 1.0	Rev. 1.0	2012. 03. 15

1. Summary

1.1. Introduction

ML-Ⅱ-DB96T is a digital input/output module that supports communication protocol for Mechatrolink-Ⅱ (MLⅡ) and uses MLⅡ communication protocol to communicate in both decentralized control component and real-time.

It includes the input/output function for controlling the input and output of digital signal as the Slave module for MLⅡ Master Card, and composed of a total of 96 terminals of input and output that can be selected for either input or output.

1.2. Application

ML-Ⅱ-DB96T module uses the upper controller and MLⅡ communication protocol, interfaces the external digital input and output signals, allowing factory automation and machine tool control, lab test, educational and other applications.

- ▶ Digital Monitoring
- ▶ Signal Forwarding
- ▶ Automated Testing Device
- ▶ Device Monitoring
- ▶ Device Interface
- ▶ Data Acquisition of Sensor Detection Device
- ▶ Industrial Digital Input and Output Control
- ▶ Monitoring the 24V Signal

2. ML-II-DB96T Specification and Configuration

2.1. Module Specification and Scheme

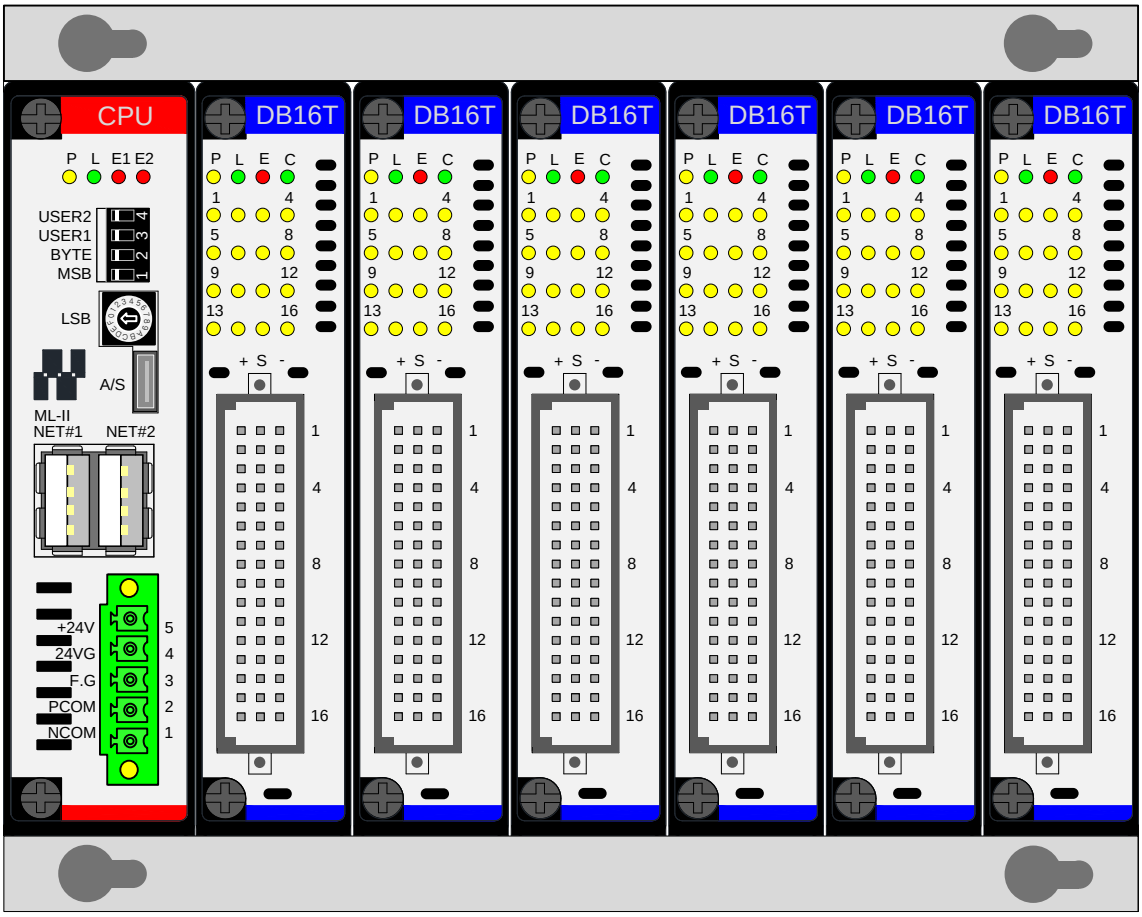


Diagram 1. ML-II-DB96T Module

Table 1. ML-II-DB96T Slave Module Specification

Item	Specification
Module Function	Digital input/output slave module with 96 terminals for Mechatrolink-Ⅱ
Node ID Change	LSB Hexadecimal Rotary Switch (0-F), MSB Switch (0-1)
Fixed Method	Fixed in four places with the Rear Panel External Corner M3 Screws
Internal Current Consumption	Max. 500mA, 24V DC (Module External TB1 Connection, I/O Power Current Consumption excluded)
LED Display	Slave CPU : Power (Yellow), Live (Green), Error1 (Red), Error2 (Red) DB16T : Power (Yellow), Link (Green), Error (Red), Com (Green) INOUT1 - INOUT96 (Yellow)
Digital Input and Output	96 Input/Output Jacks, 24V DC Level IN1~IN96 : Photo-coupler isolation (MIN. 2mA or more/Jacks) OUT1~OUT96 : Photo-coupler + Drive IC (MAX. 50mA or less/Jacks) PCOM, NCOM : Digital I/O Positive and Negative Common Connector : HIROSE (PCN10A-48P-2.54DS)
Operation Temperature and Humidity	Temperature 0 ~ 55℃ / Humidity 0 ~ 90% (No Dew Condensation)
Storage Temperature	-20 ~ 70℃
Weight	1,017g
Product Dimensions	151 (H) mm X 121 (D) mm X 100 (W) mm

2.2. ML-II-DB96T Size and Configuration

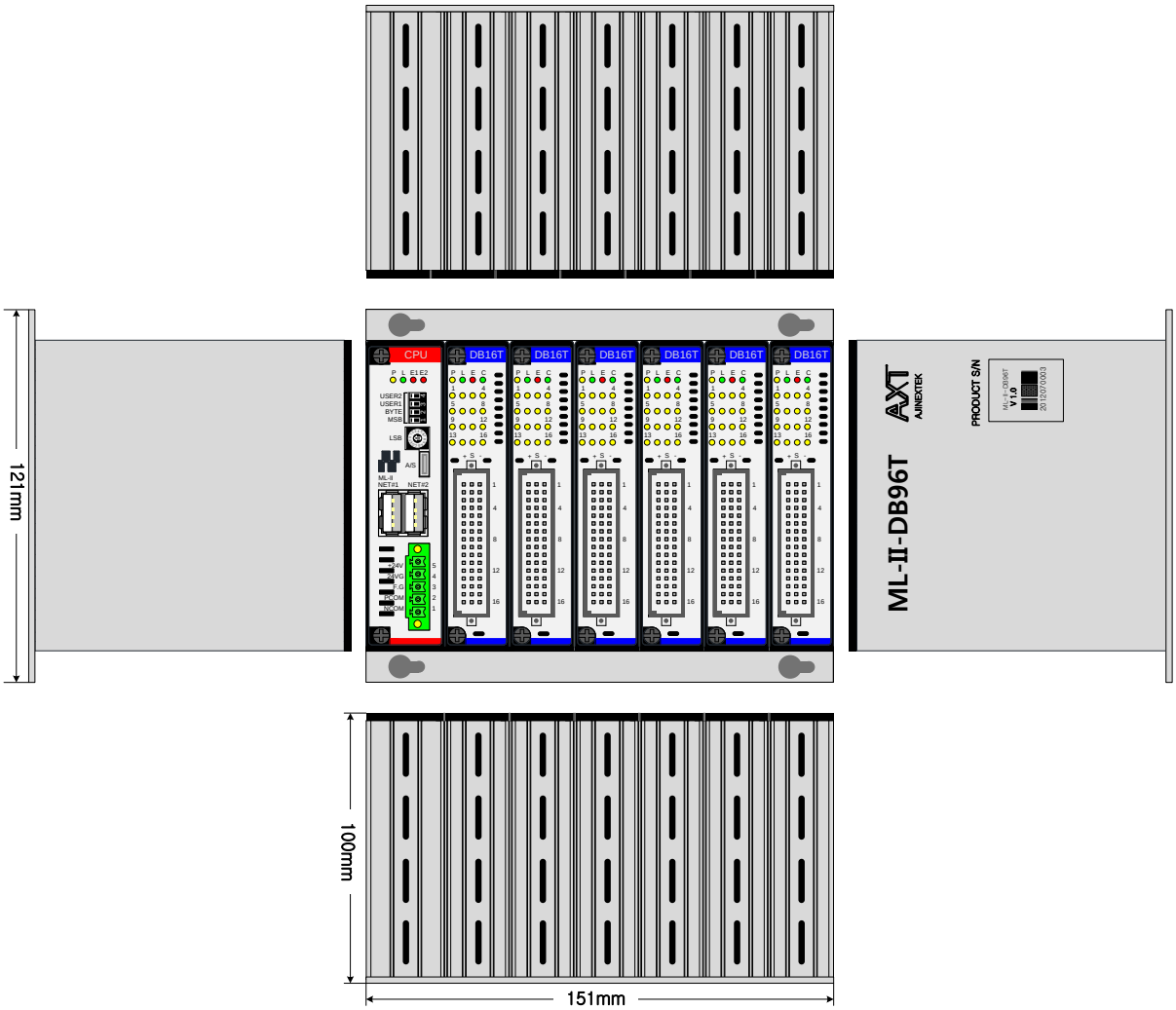


Diagram 2. ML-II-DB96T Module Size and Configuration

3. Installation

3.1. Safety and Caution

For the efficient and safe use of product, please be fully aware of the following.

3.1.1. Warnings

- In case of violating the instructions, there may be a situation involving death and serious injury.
- Do not assemble the product while the power is plugged into the wall. This is a fire hazard, and may lead to electrocution and malfunction.
- Do not work on the wiring while the power is plugged. This may lead to electrocution and malfunction.
- Ensure not to insert or allow any metallic foreign substance into the product. This is a fire hazard, and may lead to electrocution, and malfunction.
- Ensure to have the electricity technician work on the wiring. This is otherwise a fire hazard, and may lead to electrocution, and malfunction.

3.1.2. Cautions

- In case of violating the instructions, there may be a situation involving minor injury, product damage, or loss..
- Ensure to work on the wiring using the product's rated voltage and current. This is a fire hazard, and may lead to electrocution, and malfunction.
- Ensure to avoid an environment with temperature of 55℃ or higher.
- Ensure to avoid an environment with direct sunlight or exposure to sunlight. This is a fire hazard, and may lead to electrocution, and malfunction.
- Ensure to avoid an environment with room humidity of 85% or higher.
- Ensure to avoid installing the product near any flammable materials. This is a fire hazard.
- Ensure to avoid installing the product in a place with vibration. This is a fire hazard and may lead to electrocution.
- Do not dissemble, repair, or modify the product without the professional after-service technician.
- Ensure to use the product with the environmental conditions listed in the user's manual. This is a fire hazard, and may lead to electrocution, and malfunction.
- Ensure to connect only the allowed load to the outputs. This is otherwise a fire hazard, and may lead to electrocution, and malfunction.
- Ensure to avoid the external 24V power supply in constructing the interface board for the electric field. Mishandling of polarity may lead to product damage.

3.2. Installation

Install the board with the following instructions upon checking the product components.

1) Remove all static before the body contacts the ML-II-DB96T.
2) Place the product upon concerning the electric field.
3) Fix the board with four M3 bolts as shown in the diagram below.
4) Check the product has been properly installed. Check that ML-II-DB96T has been osculated and installed.

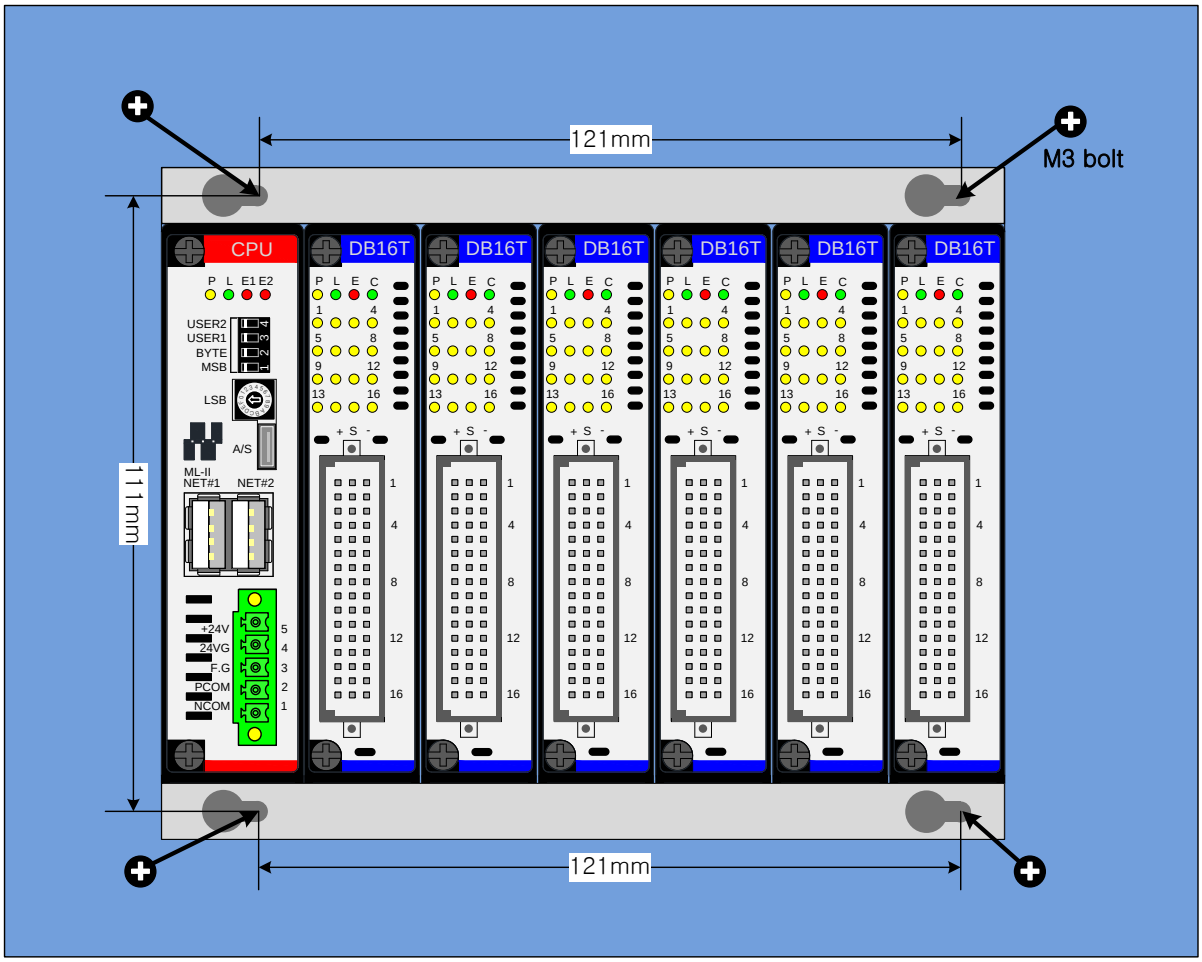


Diagram 3. ML-II-DB96T Installation

5) Operate the system upon connecting the ML-II cable to the ML-II-DB96T
6) Check that the CPU's "P" lights on and "L/E1/E2" blinks for three times in sequence.
7) Check for the proper recognition of system in the Ezsoftware using scan
8) Check for the ML-II-DB96T CPU "P" LED lighting on, "L" LED blinking, and DB16T "P/L/C" LED lighting on.

3.3. Connector Connection Information

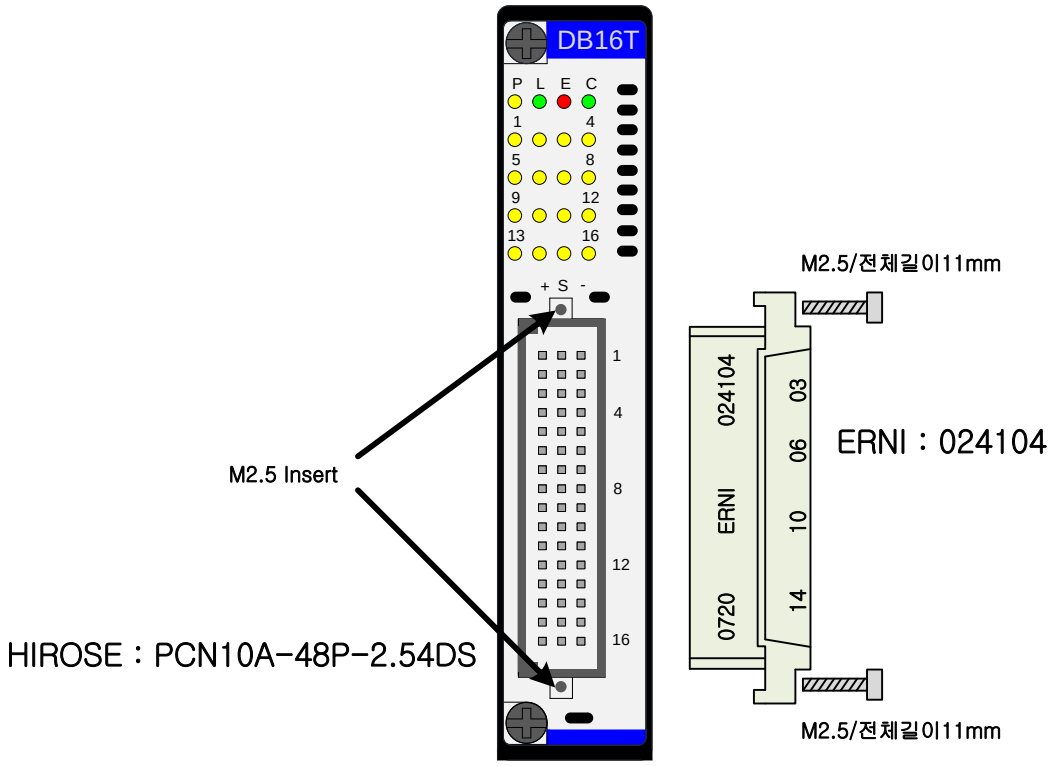
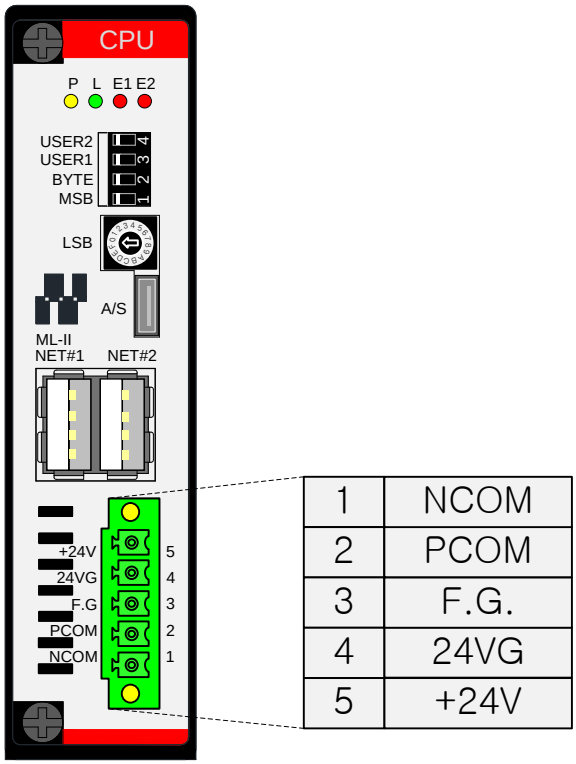


Diagram 4. ML-II-DB96T Connector Connection Information

HIROSE PCN10A-48P-2.54DS is used as the digital input and output connector. As a counterpart, ERNI 024104 may be used. In order to fix the 024104 connector, M2.5 bolt must be used, and total length is 11mm (+/- 1mm).

4. ML-II-DB96T Module Connecter Pin Configuration and Information

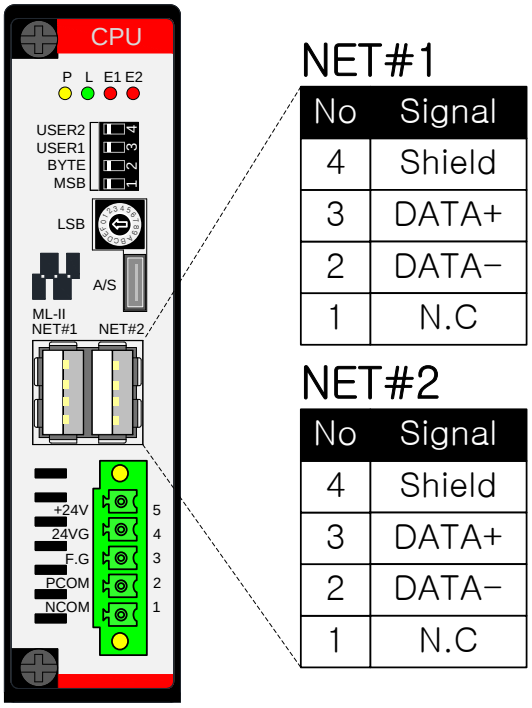
4.1. Slave CPU Connector (TB1) Pin Configuration and Information



Digram 5. ML-II-DB96T Module Power Connector (TB1) Pin Configuration

The diagram shown above is the ML-II-DB96T Module Power Input Connector (TB1). Power Input connect consists of PHOENIX MC 1,5/5-GF-3,5, a total of 5 pins. Input power uses 24V DC. The product installed with counter voltage prevention diode inside in order to ensure that the module does not supply the power to prevent operation in case the user installs the wiring in reverse.

4.2. SlaveCPU ML-II Communication Connector (NET#1, NET#2) Pin Configuration and Information



NET#1

No	Signal
4	Shield
3	DATA+
2	DATA−
1	N.C

NET#2

No	Signal
4	Shield
3	DATA+
2	DATA−
1	N.C

Digram 6. ML-II-DB96T ML-II Communication Connector (NET#1, NET#2) Pin Configuration

The diagram shown above is the Connector NET 2 of ML-II-DB96T for communication with Mechatrolink-II. The connection does not require installation in order. STP Cable with sole purpose for the communication must be used as communication cable. The minimum length of cable between nodes is 0.5M, and the maximum length of cable for the Network Component Cable is 50M per NET. The connector may look as a simple USB cable, but the lock method is required from the cable, therefore cable exclusively for communication is necessary.

⚠	Motor control of Mechatrolink-II communication method transmits and receives data in high speed through the communication cable. This method operates motor control in basis for the transmitted data; hence high reliability in the communication line is required. Therefore, irregular communication errors may occur with the use of unapproved communication cable. With this, the traffic jam or control failure or motor may occur and lead to the equipment damage or injury to the user.



Diagram 7. Specification of Cable and Connector for ML-II

※ Caution: Regular USB cable is not operational with the equipment.

4.3. SlaveCPU NODE ID Switch (LSB) Information

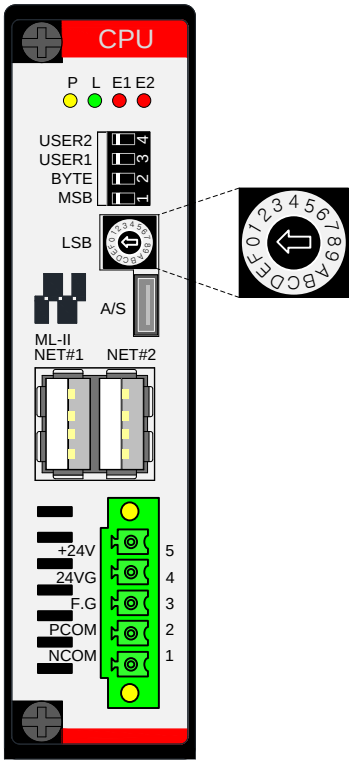


Diagram 8. ML-II-DB96T NODE ID Switch

The diagram above shows the Node Address of NODE ID Switch LSB for ML-II communication located in the Slave CPU of ML-II-DB96T module. NODE ID Switch LS can be set in a range of 0-F.

This module must be set an ID that is different from the IDs of other modules connected in the same group with ML-II-DB96T. In order to make the proper setting, use the Rotary Switch by turning the switch.

Table 2. NODE ID Setting

SWITCH		Node Address
MSB	LSB	
0	0	Not used
0	1	1
0	2	2
0	3	3
0	4	4
0	5	5
0	6	6
0	7	7
0	8	8
0	9	9
0	A	10
0	B	11
0	C	12
0	D	13
0	E	14
0	F	15
1	0	16
1	1	17
1	2	18
1	3	19
1	4	20
1	5	21
1	6	22
1	7	23
1	8	24
1	9	25
1	A	26
1	B	27
1	C	28
1	D	29
1	E	30
1	F	Not used

4.4. SlaveCPU PIANO Switch Information

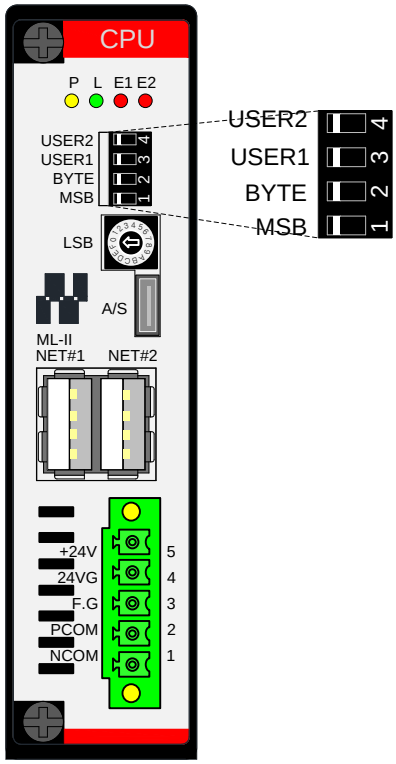


Diagram 9. ML-II-DB96T USER Setting Switch

As shown in the diagram above, the PIANO Switch located in the SlaveCPU of ML-II-DB96T module can be set to MSB, BYTE, USER1, USER2.

- MSB: Maximum of 30 slave modules are operational with ML-II, 15 are set from 0-F with the NODE ID Switch LSB and MSB sets from 0-1 for up to 30 supporting the Node Address.
- BYTE: Supports the BYTE Mode
- USER1, USER2: No set value. Allows the user to set the value.

As shown in the diagram below, ON setting is made by moving the switch from the left to right and OFF setting is made by moving the switch from the right to left.

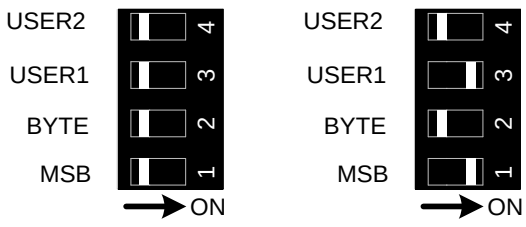


Diagram 10. Left shows ALL ON, Right shows MSB/USER1 ON state

4.5. SlaveCPU LED Display Configuration and Information

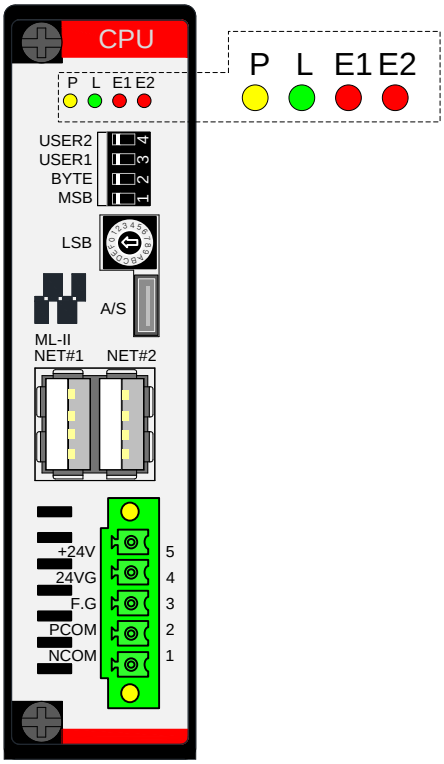


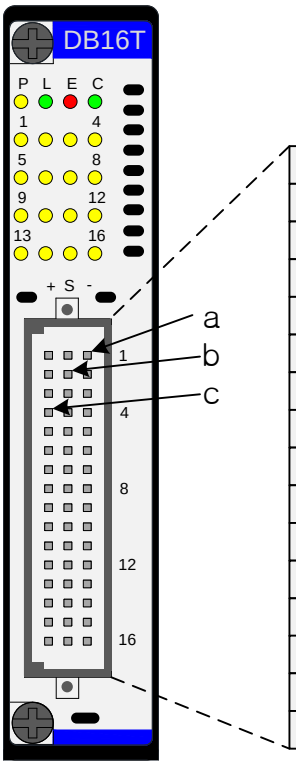
Diagram 11. SlaveCPU LED Display Panel of ML-II-DB96T Module

As shown in the diagram above, the LED Panel located in the SlaveCPU of ML-II-DB96T shows the module's status.

The LEDs showing the module status are listed below.

1) P (Power): ON if the 24V DC is supplied to the Module.
2) L (Live): ON if the communication connection cable of module is in connection with the upper controller or other slave module. Blinks if the communication is transmitted or received. Blinks slowly if there is a communication error.
3) E1 (Error1): Blinks if the communication cable is disconnected or there is a communication error after the module has been connected to the ML-II communication cable and power has been supplied. Blinks also if there is a communication error with other function module.
4) E2 (Error2): Blinks if there is a SPI communication error between the SlaveCPU and DB16 after the power has been supplied.

4.6. DB16T IO Connector Pin Configuration and Information



c1	PCOM	b1	INOUT1	a1	NCOM
c2	PCOM	b2	INOUT2	a2	NCOM
c3	PCOM	b3	INOUT3	a3	NCOM
c4	PCOM	b4	INOUT4	a4	NCOM
c5	PCOM	b5	INOUT5	a5	NCOM
c6	PCOM	b6	INOUT6	a6	NCOM
c7	PCOM	b7	INOUT7	a7	NCOM
c8	PCOM	b8	INOUT8	a8	NCOM
c9	PCOM	b9	INOUT9	a9	NCOM
c10	PCOM	b10	INOUT10	a10	NCOM
c11	PCOM	b11	INOUT11	a11	NCOM
c12	PCOM	b12	INOUT12	a12	NCOM
c13	PCOM	b13	INOUT13	a13	NCOM
c14	PCOM	b14	INOUT14	a14	NCOM
c15	PCOM	b15	INOUT15	a15	NCOM
c16	PCOM	b16	INOUT16	a16	NCOM

Diagram 12. DB16T IO Connector Pin Configuration of ML-II-DB96T.

As shown in the diagram above, IO Connector of ML-II-DB96T Module has a total of 16 terminals for input and output. DB16T supports a total of 96 terminals arranged in groups of 6.

HIROSE PCN10A-48P-2.54DS, the I/O Connector in use, is arranged into three groups: a, b, and c. A, B, and C consists of NCOM, INOUT, PCOM, respectively. The +, S, and – located in the upper part of connector are PCOM, INOUT, and NCOM, respectively. ML-II-DB96T has six DB16Ts and allows up to 96 digital input and output terminals for either input or output. In other words, a total of 96 terminals can be used in a manner where 48 terminals are used for input and other 48 terminals are used for output. Of importance for proper operation of digital input and output, positive terminal for PCOM and negative terminal NCOM must be used with the IO 24V DC power.

4.7. DB16T LED Display Configuration and Information

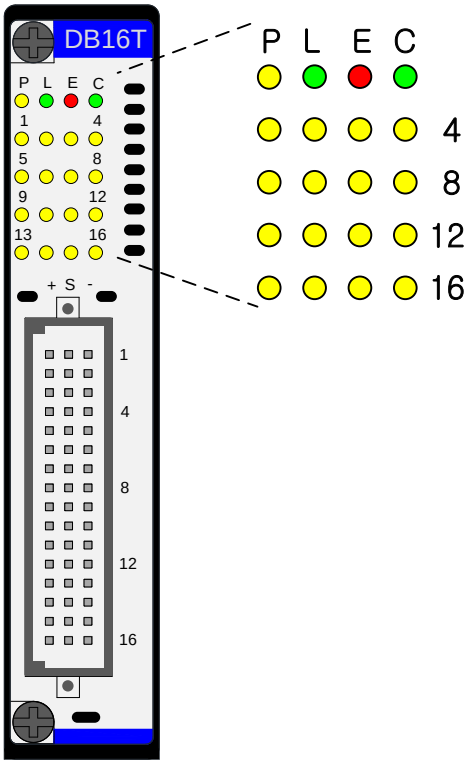


Diagram 13. DB16T LED Display Panel of ML-II-DB96T

As shown above, the LED Display Panel located in the ML-II-DB96T Module has a total of 20 LEDs. The top four LEDs display the status of DB16T, bottom 16 LEDs display the status of terminals.

The LEDs showing the module status are listed below.

1) P (Power): ON if 3.3V DC Power is supplied to the DB16T
2) L (Link): ON if the CPU is properly connected
3) E (Error): After the power is supplied,
4) C (Communication): ON if the ML-II communication is normal

The LED displays for Digital Input and Output are different for input and output.

1) For input: ON if the input is set to "L", OFF if it is set to "H"
2) For output: ON if the output is set to "L", OFF if it is set to "H"

5. ML-II-DB96T Functions

5.1. DIGITAL Input and Output Circuit Configuration and Information

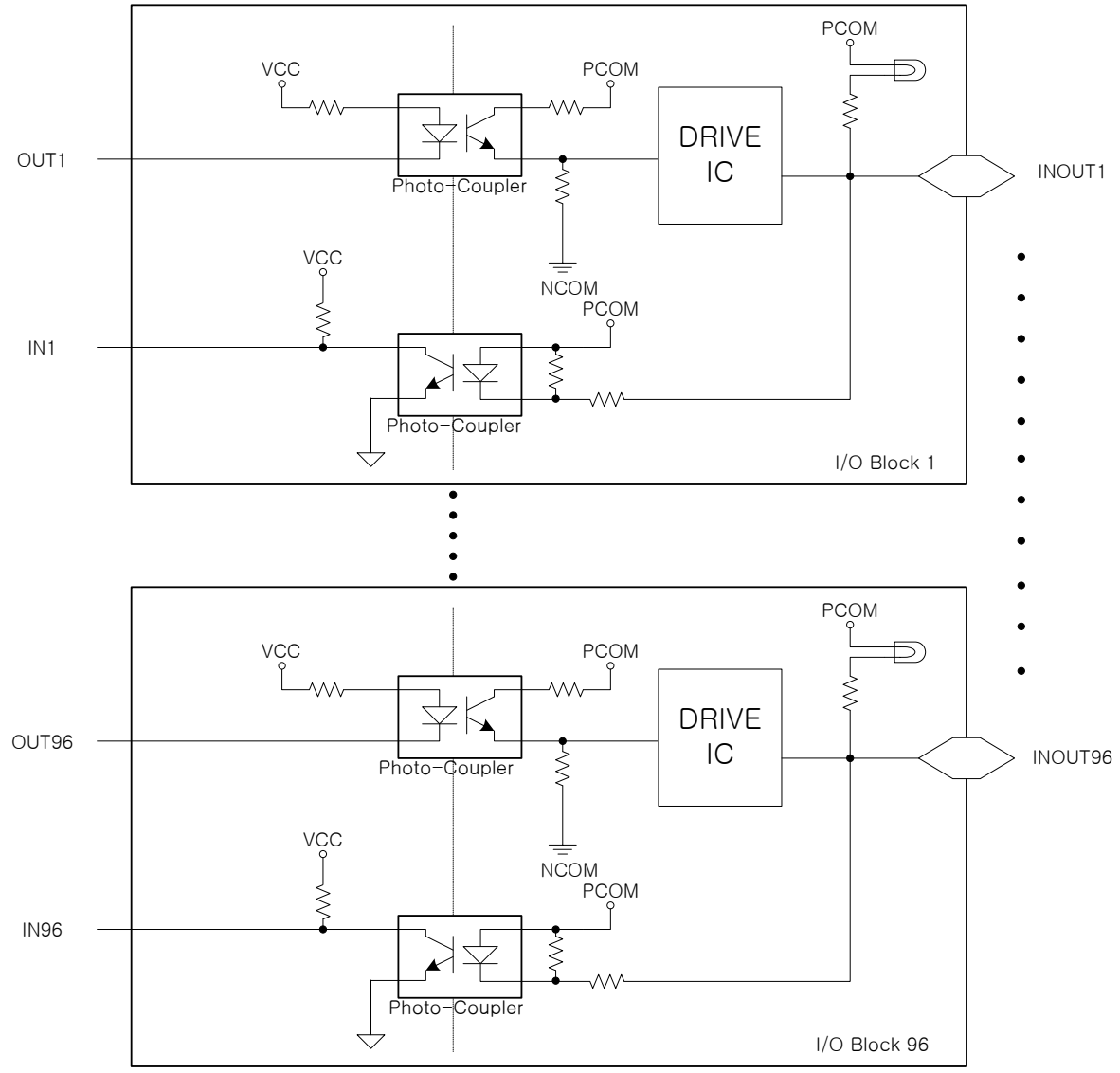


Diagram 14. DIGITAL Input and Output Circuit Configuration of ML-II-DB96T

As shown above, the ML-II-DB96T Module consists of 96 Digital Input and Output Circuit Blocks. Photo coupler insulates the circuit from external signals. In order to operate the photo coupler circuit, the PCOM (Positive Common) and NCOM (Negative Common) must be connected to the Input and Output Power of 24V DC.

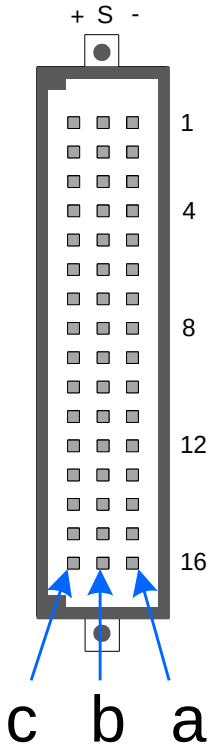
The circuit must be composed to have current at a minimum of 2mA for input. Input signal with less current will not be received as an input signal.

The circuit must be composed to have current at a maximum of 50mA for output. Output

signal with more current will operate the Protection Function of Drive IC, leading to stop the sensor operation.

The Table below shows the Pin Configuration of ML-II-DB96T IO Connector.

Table 3. ML-II-DB96T IO Connector Pin Configuration



Pin No	Pin Name	Pin No	Pin Name	Pin No	Pin Name
c1	PCOM	b1	INOUT1	a1	NCOM
c2	PCOM	b2	INOUT2	a2	NCOM
c3	PCOM	b3	INOUT3	a3	NCOM
c4	PCOM	b4	INOUT4	a4	NCOM
c5	PCOM	b5	INOUT5	a5	NCOM
c6	PCOM	b6	INOUT6	a6	NCOM
c7	PCOM	b7	INOUT7	a7	NCOM
c8	PCOM	b8	INOUT8	a8	NCOM
c9	PCOM	b9	INOUT9	a9	NCOM
c10	PCOM	b10	INOUT10	a10	NCOM
c11	PCOM	bB1	INOUT11	a11	NCOM
c12	PCOM	b12	INOUT12	a12	NCOM
c13	PCOM	b13	INOUT13	a13	NCOM
c14	PCOM	b14	INOUT14	a14	NCOM
c15	PCOM	b15	INOUT15	a15	NCOM
c16	PCOM	b16	INOUT16	a16	NCOM

5.2. Power Connection Circuit Configuration and Information

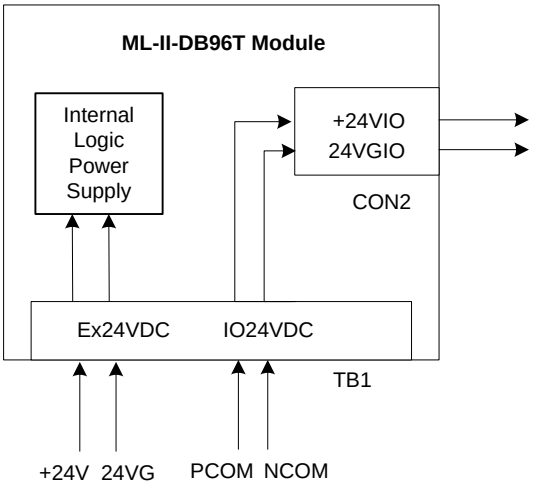
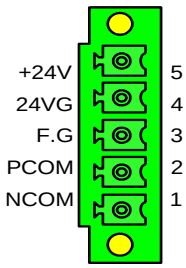


Diagram 15. Power Circuit of ML-II-DB96T Module

As shown in the diagram above, the ML-II-DB96T has divided the control power into the internal control power of 24V DC and Input/Output Control Power of 24V DC.

The Table below shows the Connector Pin Configuration of ML-II-DB96T TB1.

Table 4. ML-II-DB96T TB1 Pin Configuration



Pin No	Pin Name	Pin Description
1	NCOM	Negative Common
2	PCOM	Positive Common
3	F.G	Frame Ground
4	24VG	Control Power 24V GND
5	+24V	Control Power +24V DC

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